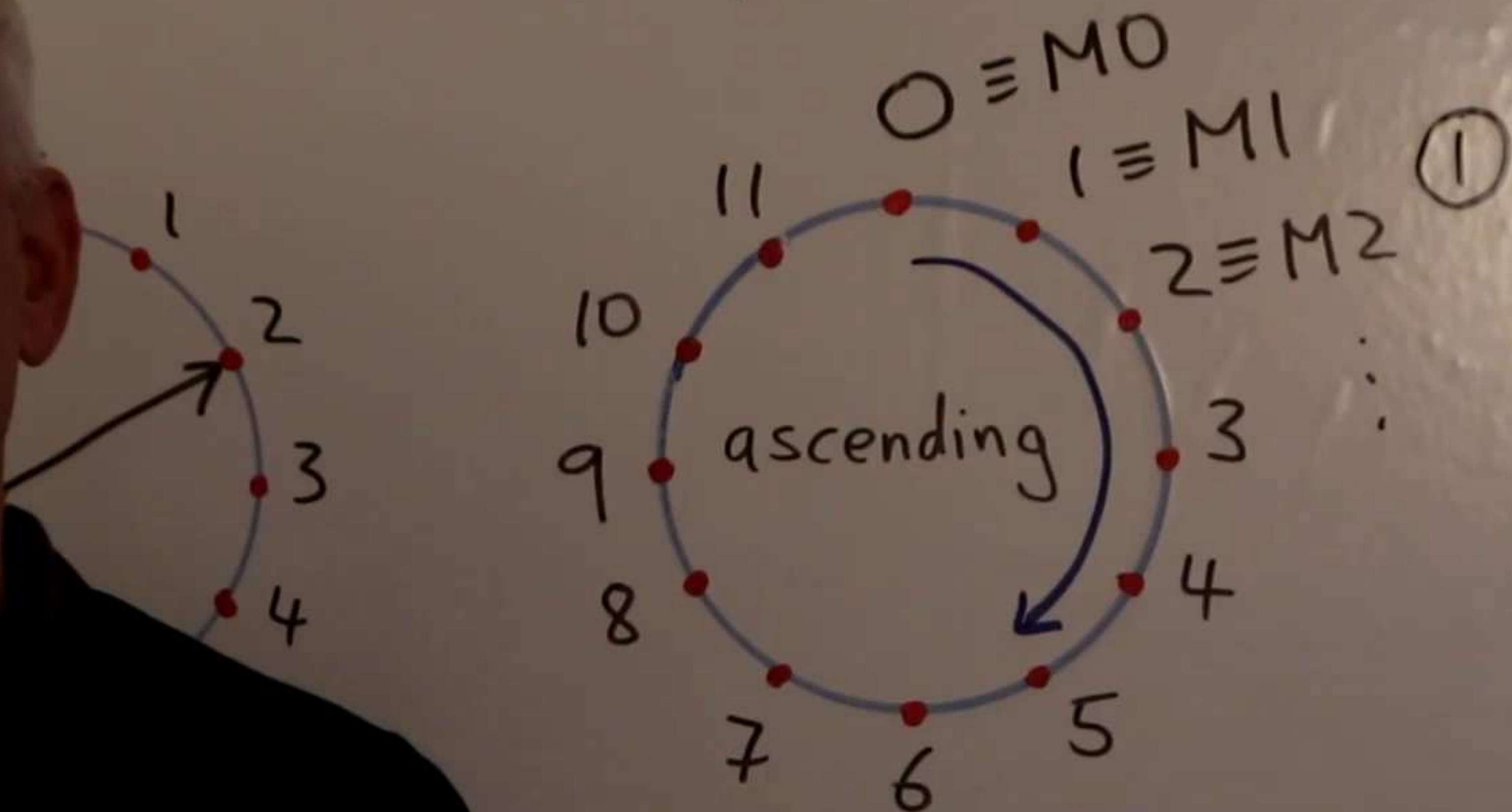


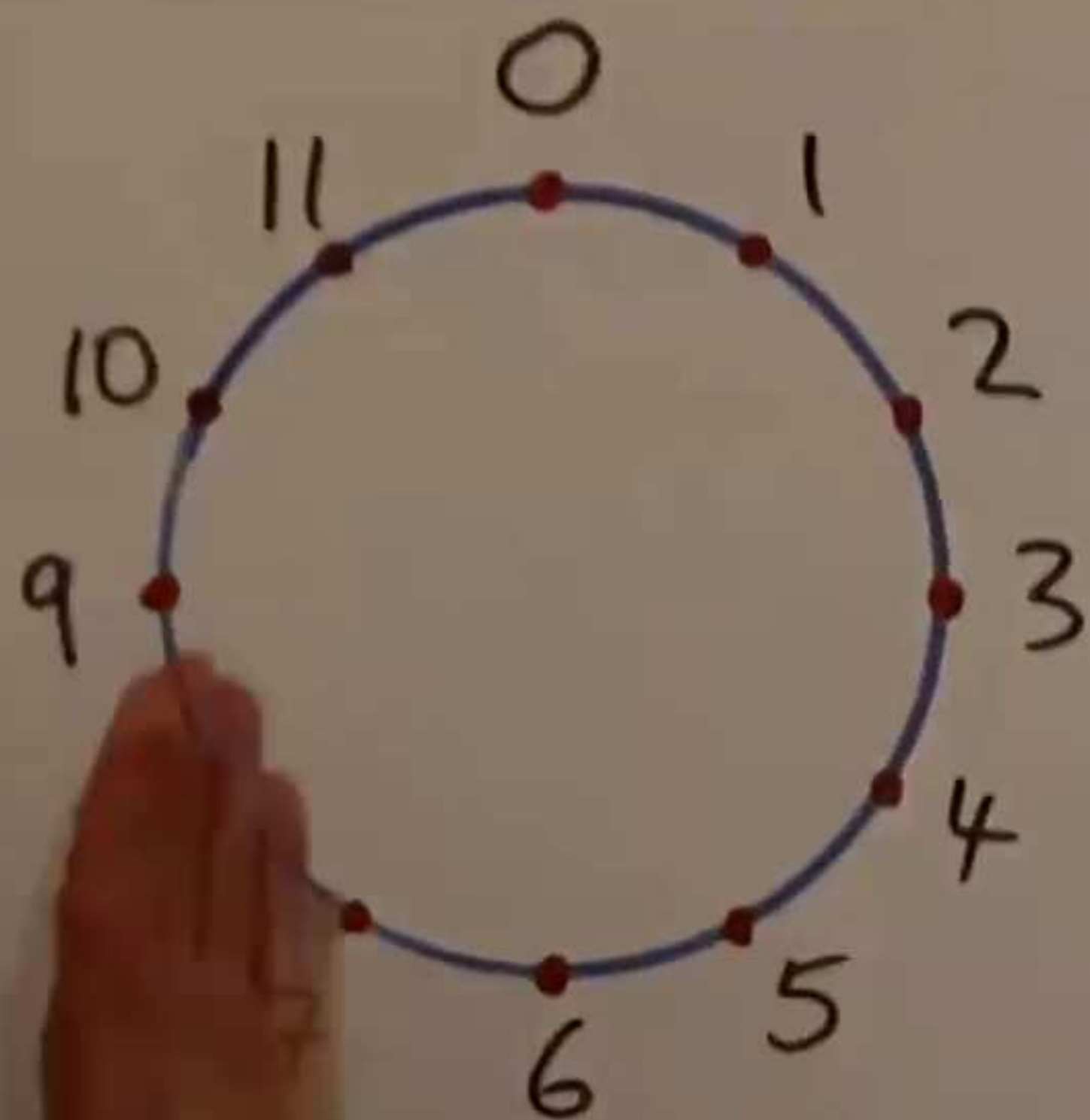
Clock mathematics and the twelve note scale (chromatic)



Twelve note scale

0, M1, ..., M11 then back to M0

The twelve
note
chromatic
scale



②
'clock view'

mod 12 arithmetic

group theory

'piano view'

6 8 10 1 3

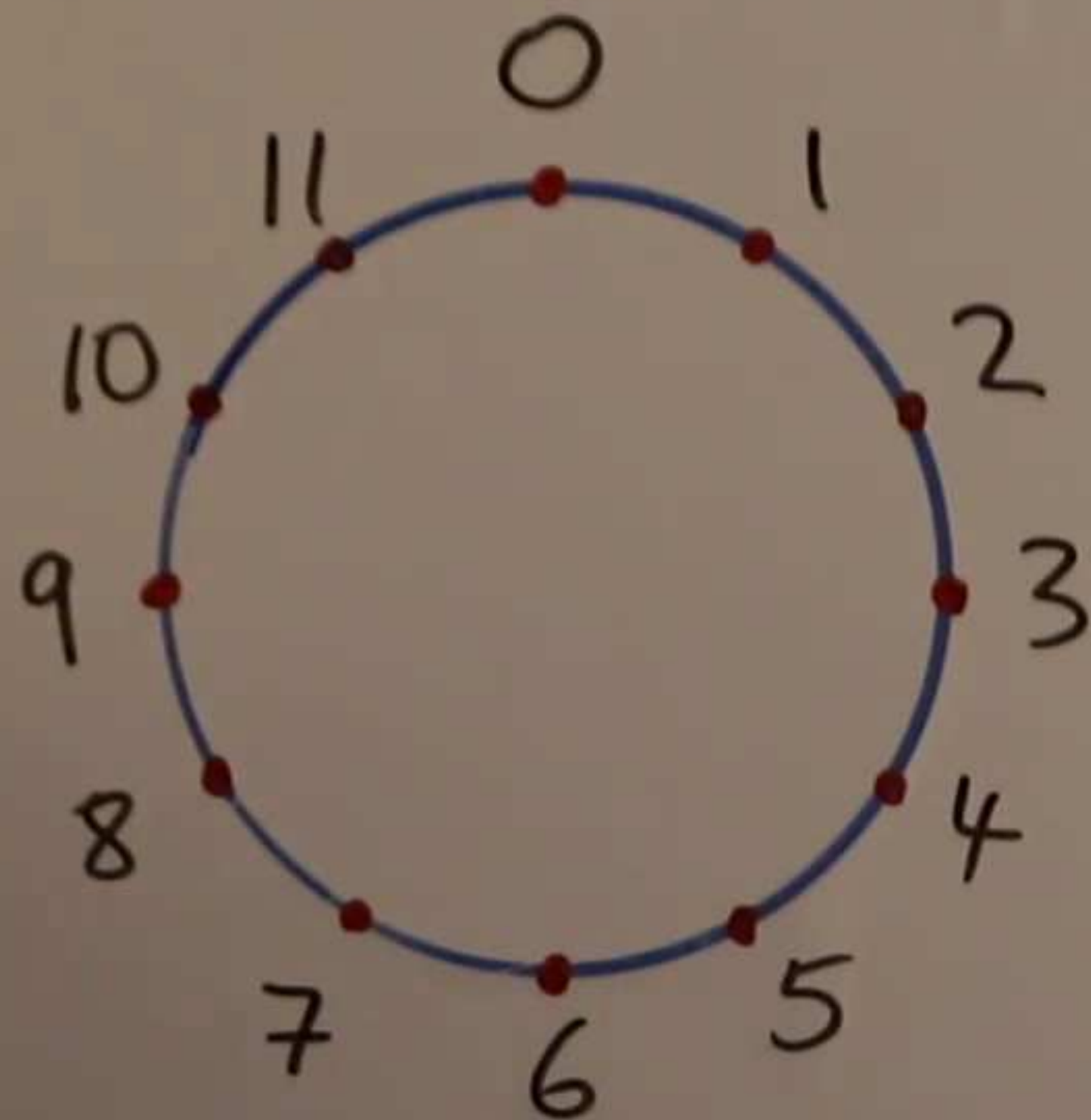


7 9 11 0 2 4

The twelve

note

tic



②

'clock view'



mod 12 arithmetic



group theory

1 3 6 8 10 1 3



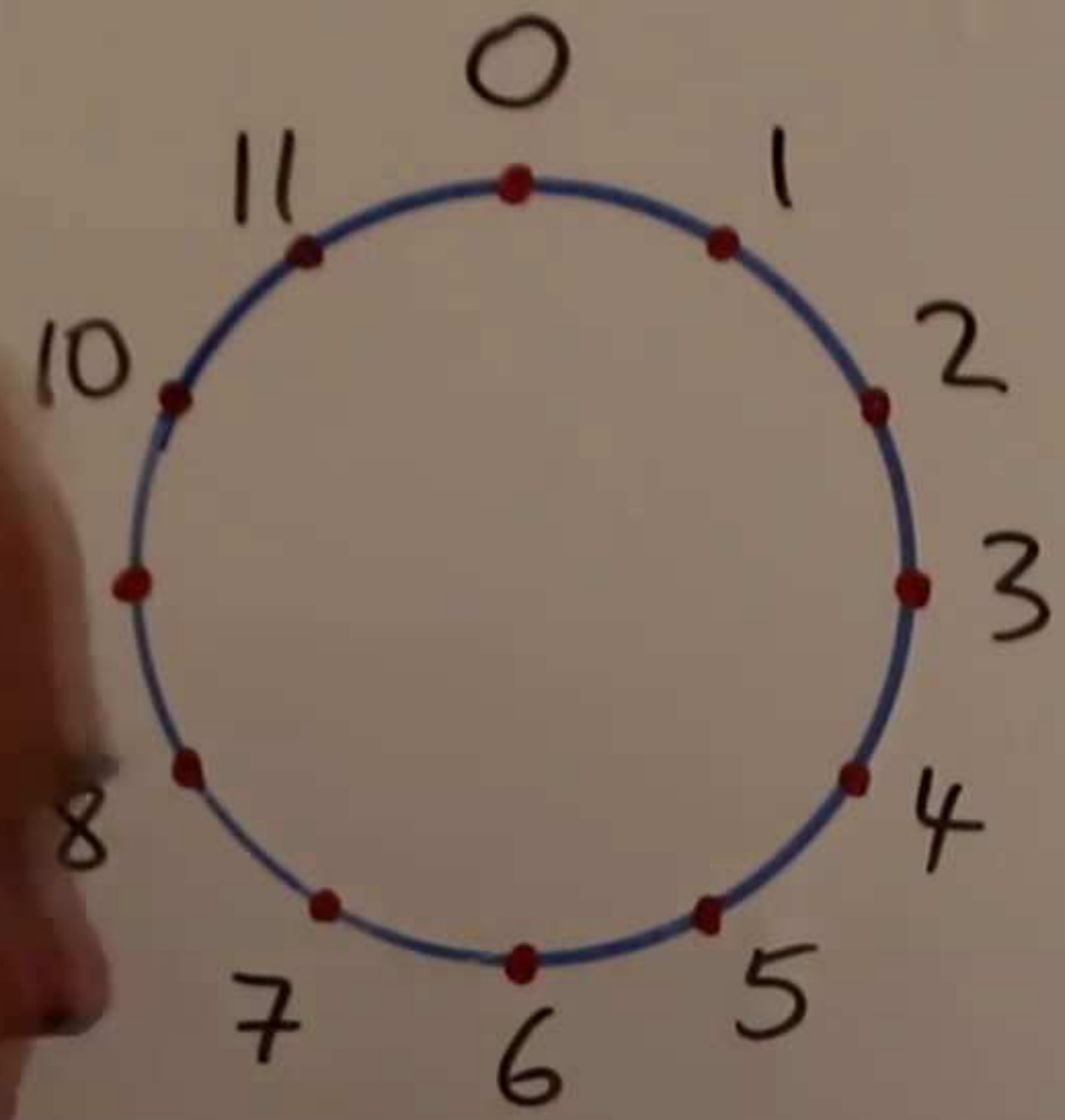
'piano view'

The twelve

note

chr

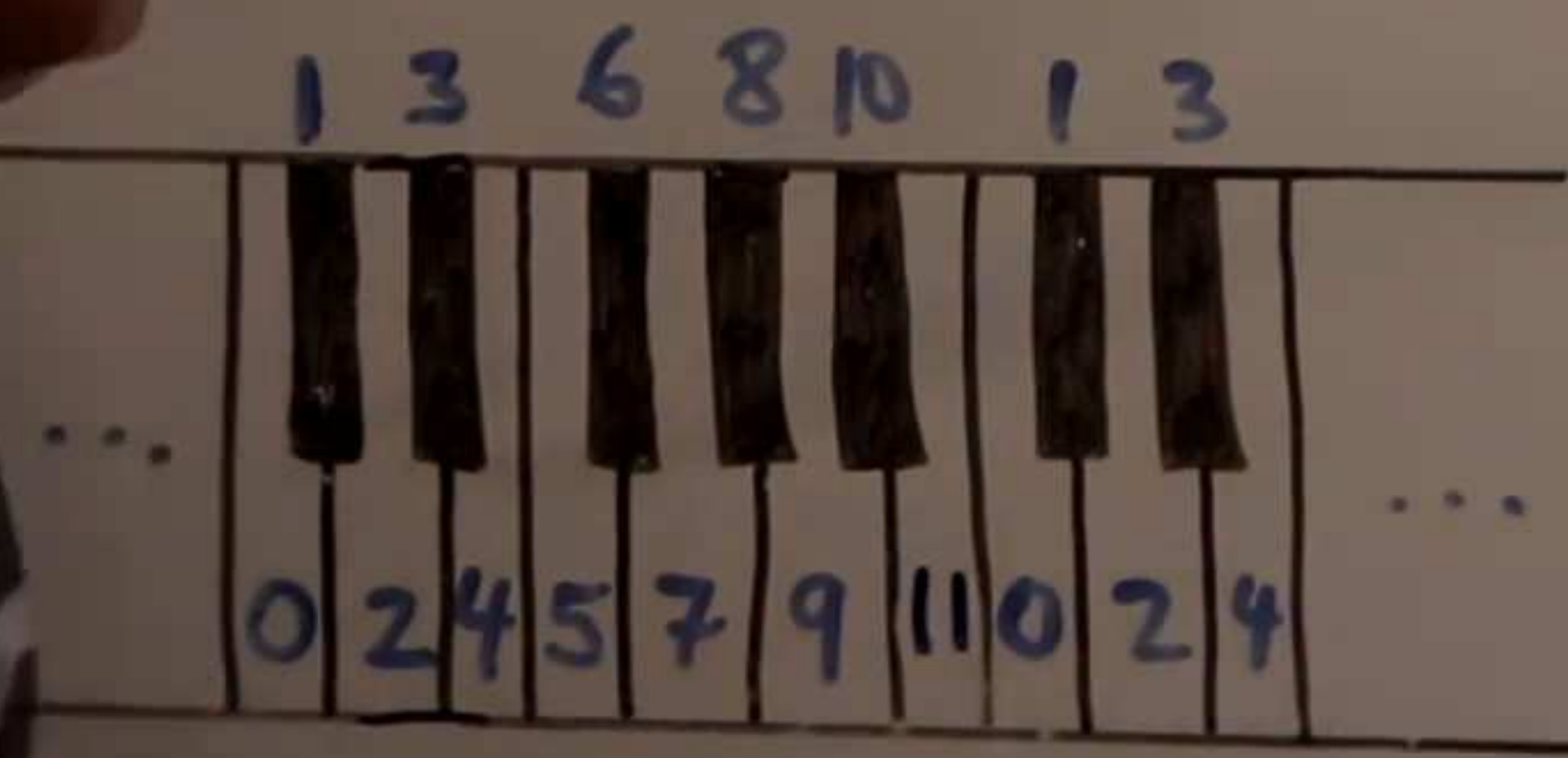
sc



'clock view'

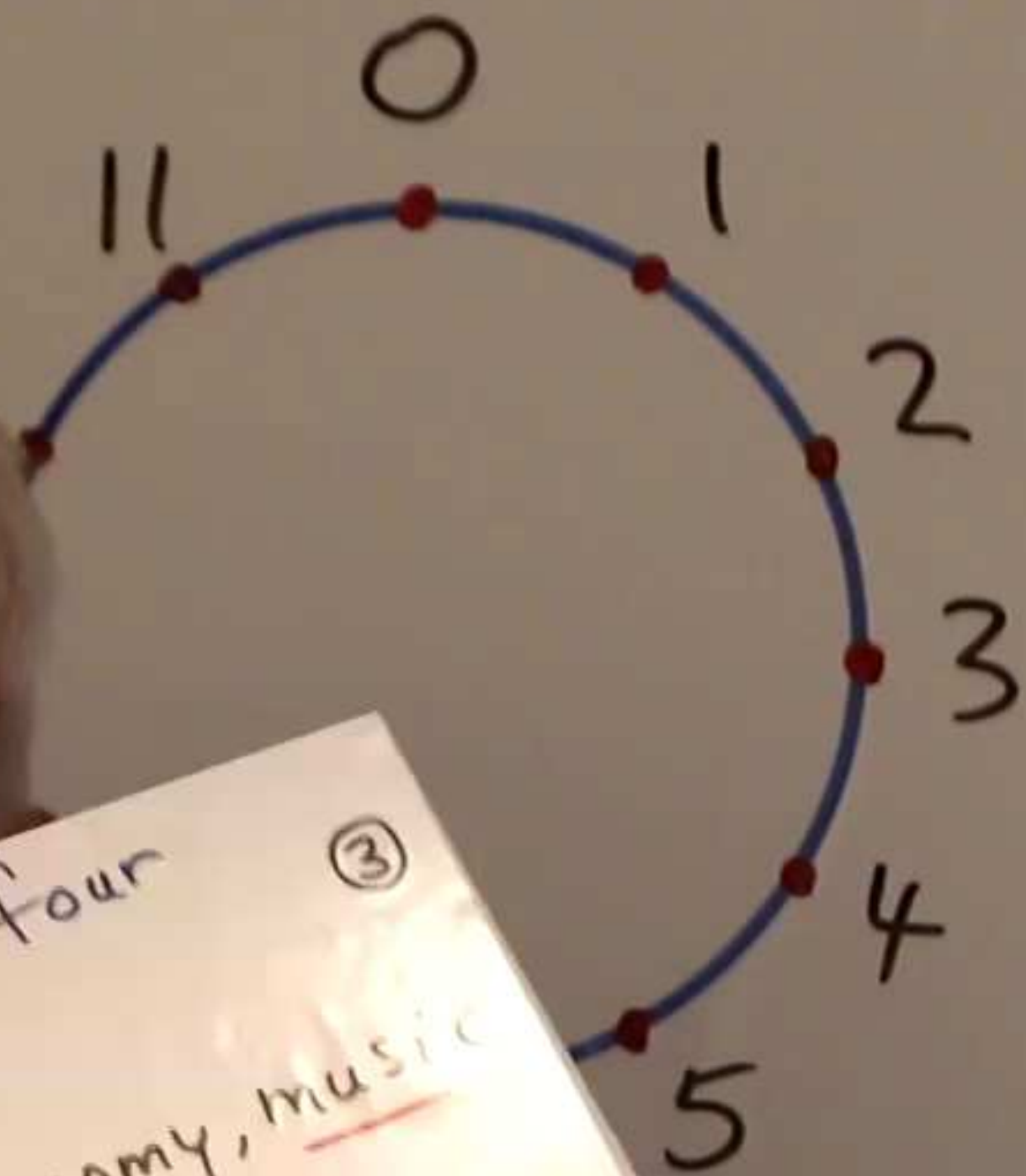
mod 12 arithmetic

group theory



'piano view'

The twelve
note
chromatic
scale



②

'clock view'



mod 12 arithmetic



group theory

'piano view'

③
Ancient Greek mathematics: four
to geometry, astronomy, music
theory, astronomy, music
theory was popular,
many mathematicians contributed
(Compendium Musicale 1618)
l'Harmonie Universelle 1636)
J. d'Alembert all wrote

Ancient Greek mathematics: four
topics (quadrivium): ③

number theory, geometry, astronomy, music

Renaissance: music theory was popular,
and later many mathematicians contributed

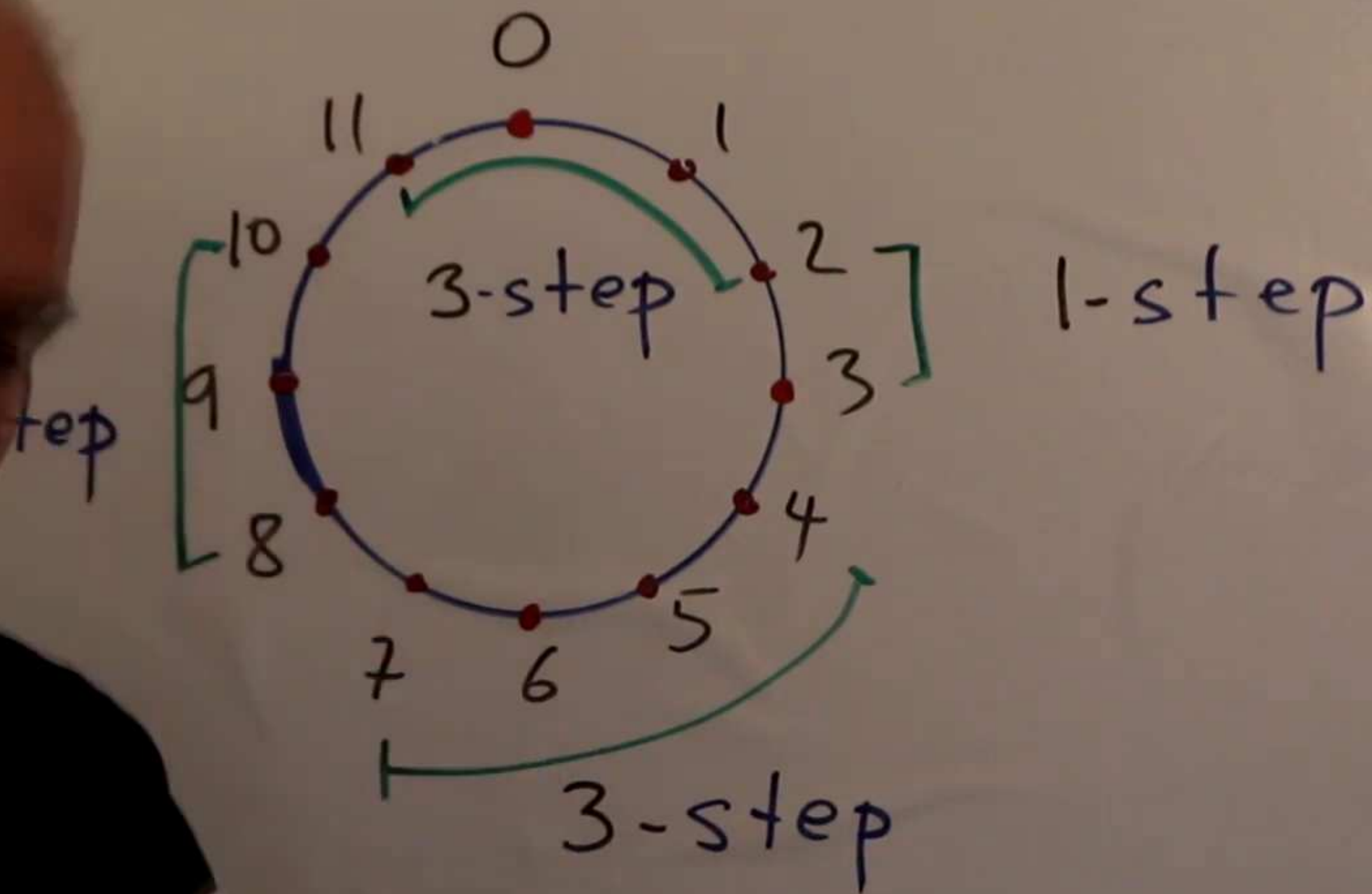
Descartes (Compendium Musicale 1618)

Mersenne (Traité de l'Harmonie Universelle 1636)

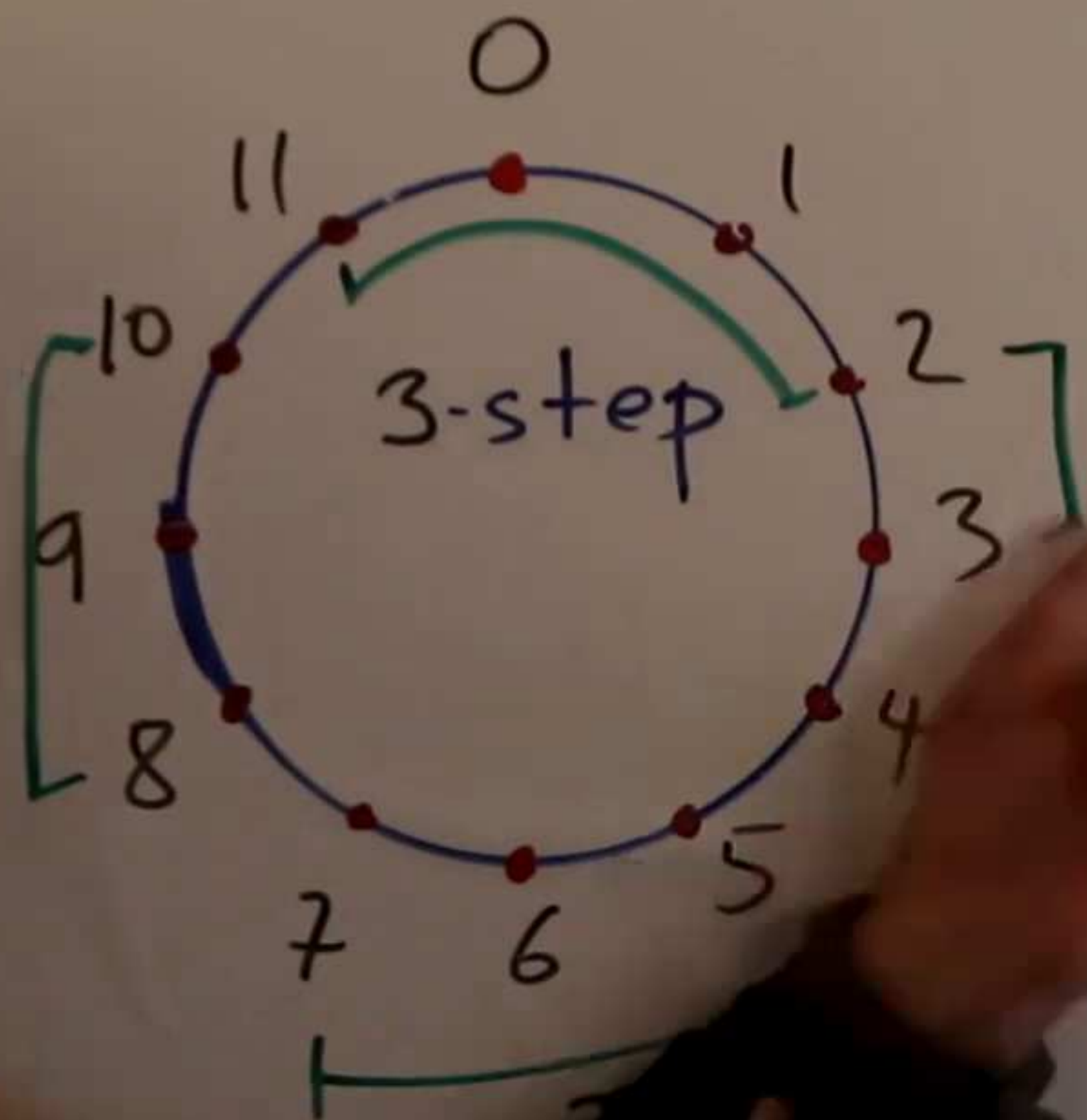
Wallis, L. Euler, J. d'Alembert all wrote

on musical theory.

Intervals : measured in steps (along
the 12 note chromatic scale)

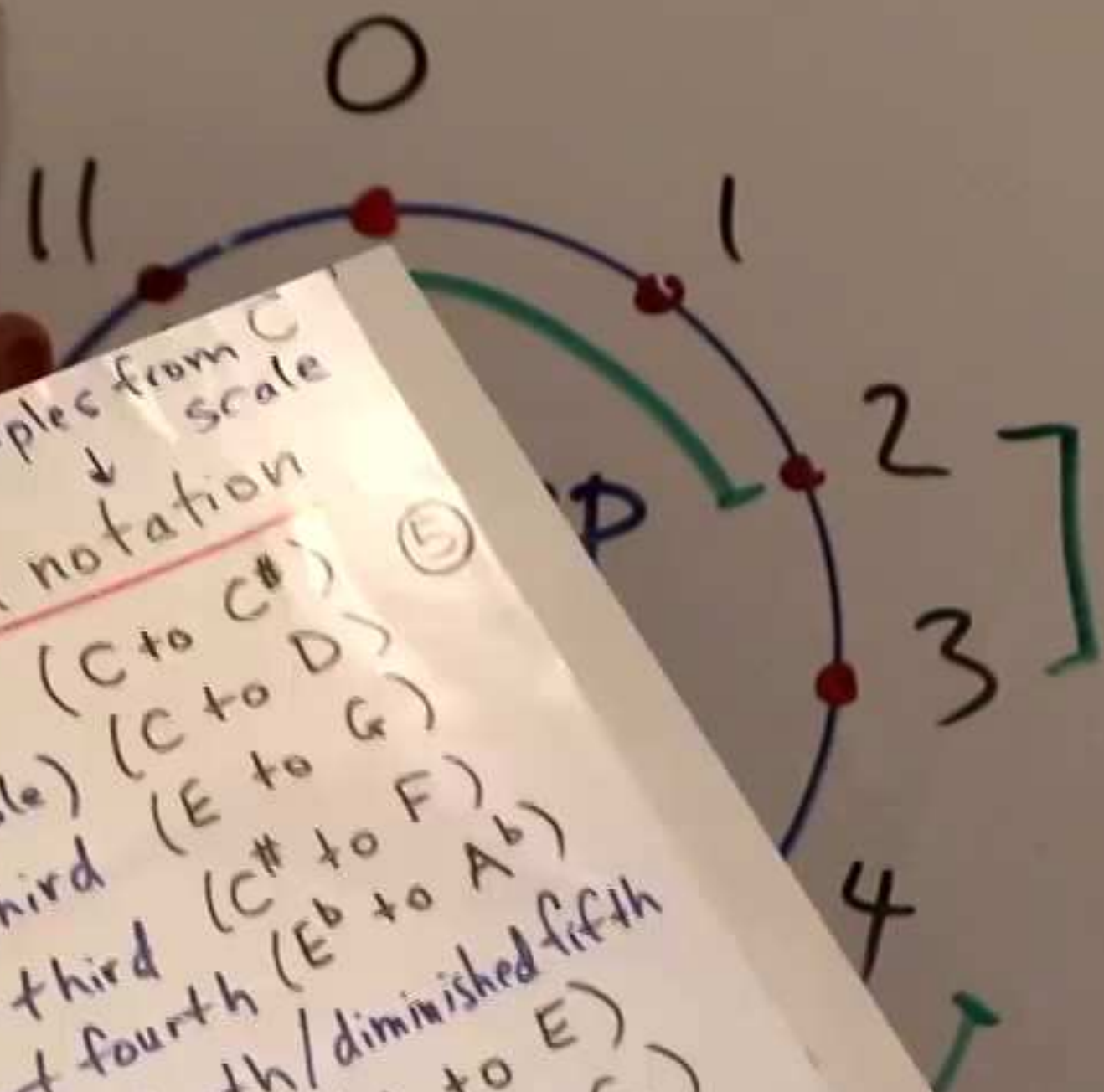


Intervals : measured in steps (along
the 12 note chromatic scale)



④

Intervals : measured in steps (along the 12 note chromatic scale)



④

1-step

Interval notation

numeric notation

step (M0 to M1)

step (M0 to M2)

step (M4 to M7)

step (M1 to M5)

step (M3 to M8)

step (M6 to M0)

Standard notation

semitone (C to C#)

tone (whole) (C to D)

minor third (E to G)

major third (C# to F)

perfect fourth (E to A)

augmented fourth/diminished fifth (E to B)

perfect fifth (A to E)

augmented fifth (B to G)

major sixth (C to A)

minor seventh (E to D)

major seventh (F to E)

Interval notation

numeric notation

step (M0 to M1)
step (M0 to M2)
step (M4 to M7)
step (M1 to M5)
step (M3 to M8)
(M6 to M0)
(M9 to M4)
(11 to M7)
(to M9)
(M2)
(M4)

Examples from C
scale
↓

Standard notation

semitone (C to C[#]) ⑤
tone (whole) (C to D)
minor third (E to G)
major third (C[#] to F)
perfect fourth (E^b to A^b)
augmented fourth / diminished fifth
perfect fifth (A to E)
augmented fifth (B to G)
major sixth (C to A)
minor seventh (E to D)
major seventh (F to E)

Interval notation

Examples from C
scale
↓

Numeric notation

- 1-step (M0 to M1)
- 2-step (M0 to M2)
- 3-step (M4 to M7)
- 4-step (M1 to M5)
- 5-step (M3 to M8)
- 6-step (M6 to M0)
- 7-step (M9 to M4)
- 8-step (M11 to M7)
- 9-step (M0 to M9)
- 10-step (M4 to M2)
- 11-step (M5 to M4)

Standard notation

- semitone (C to C[#]) (5)
- tone (whole) (C to D)
- minor third (E to G)
- major third (C[#] to F)
- perfect fourth (E^b to A^b)
- augmented fourth / diminished fifth
- perfect fifth (A to E)
- augmented fifth (B to G)
- major sixth (C to A)
- minor seventh (E to D)
- major seventh (F to E)

mod 12 arithmetic + moving along on the
12 note chromatic scale

mod 12 arithmetic (with integers) has
more relation:

$$12 = 0$$

⑥

9 10 11 0 1 2 3 4 5 6 7 8 9 10 11 0 1 2 3 4 5 6 7 8 ...

$$3 + 6 = 9, \quad 6 + 7 = 1, \quad 1 + 12 = 1,$$

$$2 \times 5 = 10, \quad 3 \times 5 = 3, \quad 4 \times 5 = 8,$$

$$-2 = 1, \quad 7 - 8 = 11, \quad 26 - 4 = 10 \quad \text{etc}$$